

Semantic Web-Enabled Intelligent VR Tutoring System for Engineering Education: A Theoretical Framework

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ABSTRACT

An unmet demand exists for personalized virtual reality (VR) training that adapts to users' abilities and performance. The primary issue lies in the high cost of setting up and maintaining virtual environments, resulting in limited flexibility for training content updates. To address this, we incorporate Semantic Web technology into VR training and explore the integration of the intelligent tutoring system. Finally, we propose a framework for a Semantic Web-Enabled Intelligent VR Tutoring System, especially for engineering education.

Index Terms: Applied computing—Education—Interactive learning environments

1 INTRODUCTION

There is an unmet need for personalized virtual reality (VR) training that adapts to users' abilities and performance. Most studies focus on the adaptation of features or feature combinations in a given training scenario, like display features and controlled elements [5], but pay less attention to the training content adaptation due to the costly setup and maintenance of virtual environments. To address this problem, it is promising to incorporate Semantic Web technology to enable ontology-driven VR training. Ontologies can facilitate the integration and management of heterogeneous knowledge like virtual three-dimensional (3D) scenes, tasks, and interactions [1]. This approach streamlines the creation and updating of training content with the support of a reasoner, though there is limited literature on this subject. An intelligent tutoring system (ITS) is a computer-based tutoring agent designed to track and analyze the student model to provide automated and personalized tutoring, including task selection, error detection and correction, step prompting, and other adaptive learning approaches [2, 3]. Integrating ITSs into VR learning environments represents a burgeoning research field; however, the current level of exploration remains limited [4].

In response to this gap, we propose a framework of Semantic Web-Enabled Intelligent VR Tutoring System, a fusion of Semantic Web technology, ITS, and VR training applications. Semantic Web technology, a knowledge representation-based artificial intelligence approach, offers an alternative pathway for implementing an ITS.

2 FRAMEWORK OVERVIEW

The Semantic Web-enabled intelligent VR tutoring system, depicted in Fig. 1, integrates a semantic layer powered by Semantic Web technology to provide intelligence. This layer comprises an ontology-based knowledge base and a reasoner, featuring several domain-specific small ontologies describing virtual worlds, user interactions, and tutoring situations in real time. These ontologies consist of domain-related taxonomies of concepts (classes), properties, reasoning rules, instances of concepts and their property assertions. The

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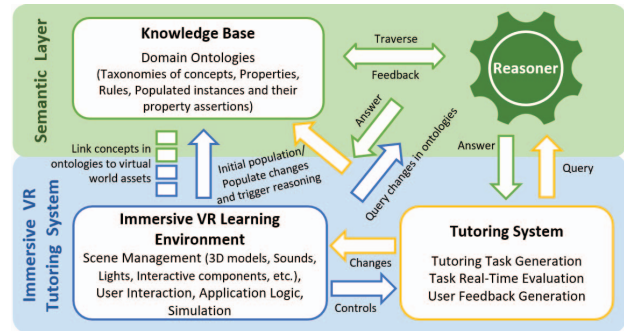


Figure 1: Framework of the Semantic Web-Enabled Intelligent VR Tutoring System (adapted from Häfner [2])

concepts are linked to virtual world assets. During system initialization, training-related object instances and their property assertions populate the knowledge base in the form of Resource Description Framework (RDF) triples, such as virtual scene instances, tutoring task instances, initial state parameters of the user and the system, etc. During tutoring, the system dynamically adds or updates instance information in response to changes in the VR environment and tutoring system, such as populating user behavior instances, updating virtual scenes or tasks, and so on.

The reasoner, a key component to implement intelligence, is activated when it receives a query from the VR environment or tutoring system or when the knowledge base receives instance updates. As the user interacts, it traverses the knowledge base data and uses ontology rules to infer necessary changes in the virtual world, or to execute prescribed tutoring actions. The application logic or tutoring system queries reasoning results to adjust the virtual world or perform further tutoring analysis. Additionally, based on functional needs, the tutoring system can also query the reasoner for necessary information in the knowledge base, such as the user's learning progress, skill state, etc.

The primary functions of the tutoring system involve generating tutoring tasks, evaluating tasks in real-time, and generating user feedback, all facilitated through interaction with the semantic layer. The outcomes of the tutoring system can change the virtual environment. For instance, task updates may change the virtual scene, and user feedback is presented on the graphical interface. The tutoring system should also provide an interactive interface in the virtual environment for users to manage their learning.

3 INSIGHTS ON IMPLEMENTATION

3.1 Build up Machine Virtual Twins

The core of a VR learning environment is the simulation of the real world. Creating a machine virtual twin involves knowledge from various domains such as electrical, mechanical, and machine programming. Embedding the semantic layer in the virtual environment not only facilitates the integration of heterogeneous data, but also by linking the semantic information to virtual 3D assets, enables automatic parameterization of simulation models and a wider range of machine behavior simulation and user interaction options.

3.2 Implementation of the Semantic Layer

OWL language is widely used for ontology modeling. Ontology rules are defined with SWRL language. SPARQL language is employed for querying and updating ontology data. Identifying all necessary domains for implementing an ITS is the premise of constructing ontologies. Here we take the ITS of a virtual flow rig (see the left side of Fig. 2) as an example, primarily for teaching the skills of controlling process parameters and operating instruments in the piping system. Most machine skill training tasks are procedural, meaning that they can be represented as a sequence of behaviors.

Based on the content and goals of the tutoring, we preliminarily determined that the semantic layer of the ITS should encompass four key domain ontologies. The first, a virtual twin ontology, serves to track the status information of components in the flow rig, with its component class hierarchy displayed in Fig. 2. The second is a behavior ontology, describing all interaction types of the flow rig (like Open, Close, Check, etc.) and their execution rules on different components. Some components have multiple interaction options. As a result, an ontology-driven simulation model is formed with the support of the reasoner and application logic, enabling automatic responses to user interactions. In addition, semantic tracking of user behavior and demonstration-based authoring of task behavior sequences can be implemented. The third, a tutoring task ontology, is used to describe procedural tasks, task evaluation results, and user feedback content. Lastly, a skill ontology is responsible for recording the learning intensity and mastery of skills, facilitating the adaptive generation of tutoring tasks.

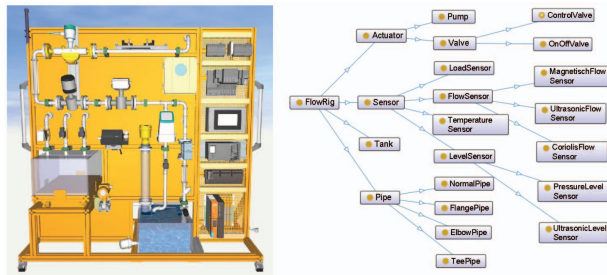


Figure 2: Flow rig virtual twin (left) and its ontology (right)

3.3 Intelligent Tutoring System

This section outlines specific requirements for implementing an ITS and provides some insights into their realization. First, the tutoring task generation module should adaptively generate the most suitable task based on the designed rules or algorithms, taking into account the student's skill level, proficiency, and forgetting curve. To assess skill levels, a straightforward method is to divide target skills into different levels, and then analyze the students' learning situation of all tasks at each level. Taking the flow rig tutoring as an example, a preliminary skill ontology is shown in Fig. 3. Skill proficiency can be assessed by formulating a performance-based penalty rule and taking the resulting task score as its indicator. Considering the knowledge forgetting curve means considering the learning intensity, that is, the learning interval. In the skill ontology, each skill is assigned proficiency and learning intensity properties, whose values are dynamically updated during tutoring and stored in the user profile at the end of tutoring. All available tasks and their corresponding scenes are pre-stored in the underlying data model of the module. The information of the newly selected task will be populated into the tutoring task ontology. A preliminary tutoring task ontology for the flow rig is shown on the right side of Fig. 3.

Secondly, the main goal of the task real-time evaluation module is to segment the user's interaction with the immersive environment, identify misbehavior, and evaluate the user's skill proficiency. In the above flow rig example, user interactions can be easily tracked

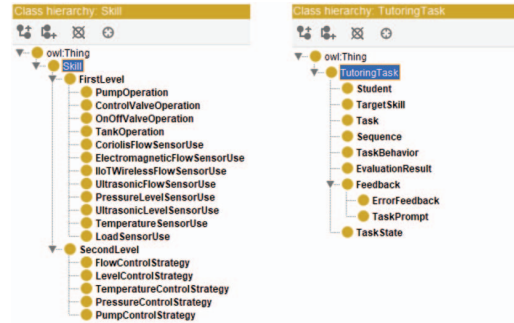


Figure 3: Skill ontology (left) and tutoring task ontology (right)

through the behavior ontology without the need for segmentation. With the task behavior sequence pre-stored in the semantic layer, the reasoner can detect misbehavior using predefined rules and then populate the results into the tutoring task ontology for generating user feedback. The module queries the detection results in time for the final evaluation of skill proficiency, which will be updated to the skill ontology.

Finally, the user feedback generation module aims to present the results of the evaluation module and task prompts to the user. To promote students' mastery of skills, the system is expected to adaptively adjust the difficulty of tasks according to the user's skill proficiency, for example, by adjusting the timing and level of detail of feedback and prompts. Based on the preset task difficulty property, the feedback and prompts are reasoned from the real-time evaluation results according to certain rules, and finally queried and output by visual or audio modules.

4 CONCLUSION AND FUTURE WORK

The proposed framework serves as a reference for implementing an intelligent VR tutoring system. The ontology-driven approach significantly reduces the cost of setting up and updating VR training scenarios. In future work, we will delve into the functional implementation of each framework module. The research results will be applied to the flow rig immersive training application as well as more complex use cases, followed by user testing and evaluation.

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